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Topic: Linux Sudo Configuration and Privilege Management

1. Create a new user account called zomatoadmin on your Linux system, and configure it so that this user can run only the apt-get update and apt-get upgrade commands using sudo without entering a password.

Hint: Edit the /etc/sudoers file using visudo and use the NOPASSWD directive for specific commands.
   - ⇨ Step 1: Create the user
     sudo useradd -m zomatoadmin
     sudo passwd zomatoadmin
     Step 2: Edit the sudoers file
     sudo visudo
     Add the following line:
     zomatoadmin ALL=(ALL) NOPASSWD: /usr/bin/apt-get update, /usr/bin/apt-get upgrade
     Step 3: Test
     Switch to the user:
     su – zomatoadmin
     Run:
     sudo apt-get update
     sudo apt-get upgrade

2. Modify the sudo privileges for your existing user so that they cannot run sudo rm -rf / but can run all other sudo commands.

Constraint: Use command-specific restrictions in the sudoers file to block only the dangerous command.
   - ⇨ Open the sudoers file:
     sudo visudo
     Assume your username is student.
     Add:

Cmnd_Alias DANGEROUS = /usr/bin/rm -rf /

student ALL=(ALL) ALL, !DANGEROUS
Test
Allowed:
sudo apt update
sudo mkdir /tmp/test
Blocked:
sudo rm -rf /

3. Simulate a scenario where a user tries to run a command with sudo that they are not allowed to. Take a screenshot of the error message and explain in 2-3 lines why the error occurs based on your sudoers configuration
   ⇨ Commands
   sudo apt-get install
   nginxguration.

4. Set up a group called myntragroup and add two users to it. Grant sudo privileges to this group so that all members can restart the networking service (systemctl restart networking) but cannot edit any files in /etc.
   ⇨ Create the group
   sudo groupadd myntragroup
   Step 2: Create two users
   sudo useradd user1
   sudo passwd user1

sudo useradd user2
sudo passwd user2
Step 3: Add users to the group
sudo usermod -aG myntragroup user1
sudo usermod -aG myntragroup user2

Verify:

id user1
Step 4: Find the correct service name
RHEL 9 usually uses NetworkManager.
Check:
systemctl list-units --type=service | grep -i network
You'll usually see:
NetworkManager.service
Step 5: Edit sudoers
sudo visudo
tep 6: Test
Allowed:
sudo systemctl restart NetworkManager

Not allowed:

sudo vi /etc/passwd

5. Use ChatGPT to generate a sample /etc/sudoers entry that allows a user called movieadmin to run all commands as root except for shutting down or rebooting the system. Copy the AI's suggestion, test it on your system, and write one line about whether it worked as expected.
   - ⇨ ChatGPT Suggestion
     Cmnd_Alias SHUTDOWN_CMDS = /usr/sbin/shutdown, /usr/sbin/reboot
     movieadmin ALL=(ALL) ALL, !SHUTDOWN_CMDS
     On some distributions, verify the command paths with
     which shutdown
     which reboot

Test
Allowed:
sudo mkdir /tmp/movie
sudo useradd demo

Blocked:

sudo reboot
sudo shutdown now

One-line observation:
The configuration worked as expected: movieadmin could run other administrative commands but was denied access to reboot and shutdown commands.